

OS RoadMap

OS Syllabus Structure

Unit	Topic	Importance
Unit 1	OS Introduction & Types	Medium
Unit 2	Process & Thread + CPU Scheduling	Very High
Unit 3	Concurrency & Synchronization	Very High
Unit 4	IPC Problems	High
Unit 5	Deadlock	Very High
Unit 6	Memory Management & Virtual Memory	Very High
Unit 7	I/O & Disk Scheduling	High
Unit 8	Unix/Linux	Medium
Unit 9	Virtualization	Low

Unit-1: Operating System Introduction — Topic Breakdown

Main Topic	Subtopics to Study	What to Focus for Exam
Introduction to Operating System	Definition of OS, Need of OS, Functions of OS	2-3 line definition + list of functions
Objectives of Operating System	Convenience, Efficiency, Ability to evolve	Short explanation of each objective
Evolution of Operating Systems	Early computer systems, Batch systems, Multiprogramming systems, Time-sharing systems	Chronological understanding
Types of Operating Systems	Batch OS, Multiprogramming OS, Time-sharing OS, Real-time OS, Distributed OS, Network OS	Definition + example
Operating System Services	Program execution, I/O operations, File system manipulation, Communication, Error detection	List with explanation
System Calls	Definition, Purpose of system calls	Concept explanation
Types of System Calls	Process control, File management, Device management, Information maintenance, Communication	Examples like fork(), exec()
Operating System Structure	Simple structure, Layered structure, Microkernel, Modular approach	Diagram based questions
Virtual Machines	Concept of virtualization, Benefits	Short explanation
Operating System Examples	Linux, Windows, UNIX, macOS	Use as examples in answers

Topics Most Likely Asked in Mid-Sem

Priority	Topic
Very High	Definition & Functions of OS
Very High	Types of Operating Systems
High	OS Services
High	System Calls
Medium	OS Structure
Medium	Evolution of OS

Unit-2: Process Management & CPU Scheduling

Main Topic	Subtopics to Study	What to Focus for Exam
Process Concept	Definition of process, Program vs Process, Components of a process	Definition + explanation with example
Process State	New state, Ready state, Running state, Waiting/Blocked state, Terminated state	Process state diagram
Process Control Block (PCB)	Process ID, Process state, Program counter, CPU registers, Memory management info, I/O status info	List and explanation of PCB fields
Process Scheduling	Concept of scheduling, Need for scheduling, Scheduling queues	Explanation + queue diagram
Types of Schedulers	Long-term scheduler, Short-term scheduler, Medium-term scheduler	Role of each scheduler
Context Switching	Definition, Steps of context switching, Overhead of context switching	Short explanation + diagram
Threads	Definition of thread, Benefits of threads	Definition + advantages
Types of Threads	User-level threads, Kernel-level threads	Comparison table
Multithreading Models	Many-to-one, One-to-one, Many-to-many	Diagrams preferred
CPU Scheduling Basics	CPU burst and I/O burst cycle, CPU scheduling criteria	Criteria explanation
Scheduling Criteria	CPU utilization, Throughput, Turnaround time, Waiting time, Response time	Definitions of each metric
CPU Scheduling Algorithms	FCFS, SJF, Priority Scheduling, Round Robin	Numerical problems + Gantt chart

Topics with Highest Exam Probability

Priority	Topic
Very High	Process State Diagram
Very High	CPU Scheduling Algorithms
Very High	Scheduling Criteria
High	Process Control Block
High	Context Switching
Medium	Threads and Multithreading Models

Unit-3: Process Synchronization

Main Topic	Subtopics to Study	What to Focus for Exam
Background of Synchronization	Concurrent processes, Need for synchronization	Concept explanation
Race Condition	Definition, Example of race condition	Clear definition + example
Critical Section Problem	Critical section concept, Structure of process, Conditions of solution	Four conditions of critical section
Critical Section Solution Requirements	Mutual exclusion, Progress, Bounded waiting	Definition of each condition
Software Solutions for Critical Section	Peterson's Algorithm, Dekker's Algorithm (if included)	Steps of algorithm
Hardware Synchronization Methods	Test-and-set instruction, Compare-and-swap	Basic explanation
Mutex Locks	Concept of mutex lock	Definition + use case
Semaphores	Definition, Semaphore operations (wait, signal)	Algorithm steps
Types of Semaphores	Binary semaphore, Counting semaphore	Difference between them
Classical Synchronization Problems	Producer-Consumer problem	Solution using semaphore
	Readers-Writers problem	Basic explanation
	Dining Philosophers problem	Diagram + explanation

Topics with Highest Exam Probability

Priority	Topic
Very High	Critical Section Problem
Very High	Semaphore (wait and signal operations)
Very High	Producer-Consumer Problem
High	Race Condition
High	Peterson's Algorithm
High	Dining Philosophers Problem
Medium	Mutex Locks
Medium	Hardware Synchronization

Unit-4: Inter process Communication (IPC)

Main Topic	Subtopics to Study	What to Focus for Exam
Interprocess Communication Concept	Definition of IPC, Need for IPC in operating systems	Clear definition + purpose
Communication Models	Shared Memory Model, Message Passing Model	Difference between both models
Shared Memory	Concept of shared memory, How processes share memory	Diagram + explanation

Message Passing System	Direct communication, Indirect communication	Definition with examples
Communication Operations	Send operation, Receive operation	Syntax and working
Direct Communication	Process-to-process communication, Naming of processes	Explanation with characteristics
Indirect Communication	Mailboxes (ports), Communication through mailbox	Concept explanation
Message Passing Issues	Blocking communication, Non-blocking communication	Difference between them
Buffering in IPC	Zero capacity buffer, Bounded buffer, Unbounded buffer	Short explanation of each
Pipes	Concept of pipe communication, Ordinary pipes	Basic working
Named Pipes	FIFO communication, Communication between unrelated processes	Explanation + example
Message Queues	Concept of message queues	Basic explanation
Shared Memory Implementation	Use of shared memory in IPC	Practical idea + explanation

Topics with Highest Exam Probability

Priority	Topic
Very High	Interprocess Communication Concept
Very High	Shared Memory vs Message Passing
Very High	Pipes and Named Pipes
High	Blocking vs Non-blocking Communication
High	Direct vs Indirect Communication
High	Buffering in Message Passing
Medium	Message Queues
Medium	Shared Memory Implementation

Unit-5: Deadlocks Topic Breakdown

Main Topic	Subtopics to Study	What to Focus for Exam
Deadlock Concept	Definition of deadlock, Example of deadlock	Clear definition + example
System Model	Resource types, Process-resource interaction	Concept explanation
Necessary Conditions for Deadlock	Mutual exclusion, Hold and wait, No preemption, Circular wait	Definition of all four conditions
Resource Allocation Graph	Graph representation of processes and resources	Diagram + interpretation
Deadlock Prevention	Methods to prevent deadlock by breaking conditions	Explanation of techniques
Deadlock Avoidance	Safe state concept, Unsafe state	Definition + explanation
Banker's Algorithm	Safety algorithm, Resource request algorithm	Algorithm steps

Deadlock Detection	Detection algorithm	Basic explanation
Recovery from Deadlock	Process termination, Resource preemption	Methods of recovery

Topics with Highest Exam Probability

Priority	Topic
Very High	Necessary Conditions for Deadlock
Very High	Resource Allocation Graph
Very High	Banker's Algorithm
High	Deadlock Prevention
High	Deadlock Avoidance
High	Safe State Concept
Medium	Deadlock Detection
Medium	Recovery from Deadlock

Unit-6: Memory Management Topic Breakdown

Main Topic	Subtopics to Study	What to Focus for Exam	Priority
Memory Management Concept	Definition, Need for memory management	Concept explanation	High
Address Binding	Compile-time, Load-time, Execution-time binding	Definitions + differences	Medium
Logical and Physical Address	Logical address space, Physical address space	Concept explanation + comparison	High
Memory Allocation	Contiguous memory allocation	Basic explanation	High
Partition Allocation	Fixed partitioning, Variable partitioning	Differences + characteristics	High
Fragmentation	Internal fragmentation, External fragmentation	Definitions + examples	Very High
Compaction	Memory compaction concept	Working explanation	Medium
Paging	Paging concept, Pages, Frames, Address translation	Diagram + working	Very High
Page Table	Structure of page table, Mapping	Diagram + explanation	High
Segmentation	Segments, Segment table, Address translation	Diagram + explanation	Very High
Segmentation with Paging	Combined segmentation and paging	Basic concept + architecture	Medium
Virtual Memory Concept	Definition, Need for virtual memory	Concept explanation	High
Demand Paging	Demand paging concept, Lazy loading	Working steps + diagram	Very High
Page Fault	Definition, Page fault handling steps	Flow explanation	Very High
Page Replacement	Need for replacement, Replacement process	Basic concept	High
FIFO Page Replacement	FIFO algorithm, Page fault calculation	Numerical problems	Very High
LRU Page Replacement	LRU algorithm, Stack concept	Numerical problems	Very High
Optimal Page Replacement	Optimal replacement algorithm	Numerical problems	Very High
Frame Allocation	Equal, Proportional allocation	Basic explanation	Medium

Thrashing	Thrashing concept, Causes, Effects	Definition + prevention	High
Working Set Model	Working set window, Locality	Thrashing control	Medium
Security Issues	Memory protection, Unauthorized access	Basic explanation	Medium

Highest Exam Probability Topics

Priority	Topics
Very High	Paging
Very High	Segmentation
Very High	Internal vs External Fragmentation
Very High	Demand Paging
Very High	Page Fault
Very High	FIFO, LRU, Optimal Algorithms
High	Logical vs Physical Address
High	Contiguous Memory Allocation
High	Fixed vs Variable Partitioning
High	Virtual Memory
High	Thrashing
Medium	Address Binding
Medium	Frame Allocation
Medium	Working Set Model
Medium	Segmentation with Paging
Medium	Security Issues

Unit-7: I/O Systems & Disk Scheduling

Topic Breakdown

Main Topic	Subtopics to Study	What to Focus for Exam
I/O Hardware	I/O devices, Device controllers, I/O ports	Basic concept explanation
I/O Structure	Communication between CPU and I/O devices	Working concept
I/O Techniques	Programmed I/O, Interrupt-driven I/O, Direct Memory Access (DMA)	Differences and working
I/O Buffering	Single buffering, Double buffering, Circular buffering	Short explanation
Disk Structure	Disk components, Tracks, Sectors, Cylinders	Basic disk layout
Disk Scheduling Concept	Need for disk scheduling	Concept explanation
Disk Scheduling Algorithms	FCFS, SSTF	Algorithm working
Advanced Disk Scheduling	SCAN, C-SCAN	Working + advantages
Disk Management	Disk formatting, Boot block, Bad block recovery	Basic explanation

Topics with Highest Exam Probability

Priority	Topic
Very High	Disk Scheduling Algorithms (FCFS, SSTF)
Very High	SCAN and C-SCAN Algorithms
Very High	Disk Structure

High	DMA (Direct Memory Access)
High	Programmed I/O vs Interrupt-driven I/O
Medium	I/O Buffering
Medium	Disk Management

Unit-8: Unix/Linux Operating System

Topic Breakdown

Main Topic	Subtopics to Study	What to Focus for Exam
Development of Unix/Linux	History of Unix, Evolution of Linux, Features of Unix/Linux	Chronological development + key features
Unix/Linux Architecture	Kernel, Shell, File System, Utilities	Architecture understanding
Role & Functions of Kernel	Process management, Memory management, File system management, Device management	Functions of kernel
System Calls	Concept of system calls, Examples (fork, exec, open, read, write)	Definition + examples
Elementary Linux Commands	Basic commands (ls, cd, pwd, mkdir, rmdir, cp, mv, rm)	Command usage
Shell Programming	Shell concept, Basic shell scripting, Variables and simple scripts	Basic explanation
Directory Structure	Root directory (/), Important directories (/bin, /etc, /usr, /var, /home)	Directory hierarchy
System Administration	User management, File permissions, Basic system monitoring	Basic admin tasks

Topics with Highest Exam Probability

Priority	Topic
Very High	Role & Functions of Kernel
Very High	System Calls
High	Directory Structure
High	Elementary Linux Commands
Medium	Development of Unix/Linux
Medium	Shell Programming
Medium	System Administration

Unit-9: Virtualization Topic Breakdown

Main Topic	Subtopics to Study	What to Focus for Exam
Virtualization Concept	Definition, Need of virtualization	Basic concept + purpose
Virtual Machines (VM)	VM concept, Working of VM	Architecture + explanation
Multiple Operating Systems on Single Hardware	Running multiple OS simultaneously	Basic workflow
Running One OS on Top of Another	Host OS and Guest OS concept	Diagram + concept
True / Pure Virtualization	Definition, complete virtualization	Characteristics
Processor Virtualization	CPU virtualization issues	Basic explanation
Memory Management in Virtualization	Virtual memory handling in VM	Basic architecture
I/O Management in Virtualization	Virtualized I/O handling	Concept explanation
VMware ESXi	VMware hypervisor platform	Features + use
Microsoft Hyper-V	Microsoft virtualization platform	Features + use
Xen Variants	Xen hypervisor concept	Basic explanation
Java Virtual Machine (JVM)	JVM architecture and role	Working concept
Linux VServer	Linux-based virtualization	Basic idea
Virtual Machine Architecture	Host machine, hypervisor, guest OS	Diagram + layers
Android Virtual Machine	Android Runtime / Dalvik VM	Basic architecture
Case Study: Linux OS	Linux virtualization support	Short explanation
Case Study: Windows OS	Hyper-V and virtualization support	Short explanation

Topics with Highest Exam Probability

Priority	Topic
Very High	Virtual Machine Concept
Very High	True (Pure) Virtualization
High	Processor, Memory and I/O Virtualization
High	Virtual Machine Architecture
Medium	VMware ESXi, Hyper-V, Xen
Medium	Java Virtual Machine
Medium	Linux VServer / Android VM